

Qingtao Liu

ML Engineer | Quantitative Software Developer

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Machine Learning Engineer with focus on NLP, large model engineering, and quantitative finance infrastructure. Proficient in RAG architecture design, LLM integration (Deepseek, Qwen), and full-stack deployment (Docker, Kubernetes). Experienced in building scalable AI systems and financial risk computation pipelines using PyTorch, C++/Python interfaces, and vector databases. Skilled in model optimization, CI/CD automation, and bridging AI theory with production applications.

EDUCATION & AWARDS

Bachelor of Applied Science and Engineering (B.A.Sc) + PEY Co-op | University of Toronto | 2022 – 2027

- Dean's Honour List – University of Toronto Faculty of Engineering, May 2023

RBC Hackathon – Built codebase LLM agent using Aiden platform to generate specialized agents from GitHub repositories; placed 6th out of 78 teams

WORK EXPERIENCE

Software Engineer (PEY Co-op) | RBC Capital Markets, Toronto | Sep 2025 – Aug 2026

- Migrating FELIX fixed income risk engine from Windows/Data Synapse to Linux/RHEA using Argo for workflow orchestration and Kubernetes for container deployment.
- Exposing C++ pricing libraries to Python via SWIG interfaces, enabling integration with Argo-based batch pipelines for overnight FRT, SRT, and XVA risk calculations.
- Deploying Kubernetes queue deployments on RHEA to spin up worker pods for concurrent FELIX risk tasks; monitoring pods via kubectl and automating CI/CD with Jenkins and Docker.

Algorithm Intern (LLM & RAG) | Sunyard Technology Co., Ltd., Hangzhou | Apr 2025 – Jul 2025

- Built RAG system (Milvus, Elasticsearch, MySQL) integrating Qwen LLM and BGE-M3 embedding for knowledge grounding and reduced hallucinations.
- Enhanced Qwen's multi-turn dialogue capability with NvN memory logic; optimized retrieval latency for real-time performance.
- Deployed Qwen applications (local + Docker), optimized serving efficiency, and designed LLM memory mechanisms.

Software Developer Engineer | Beijing Lingdang Information Technology, Beijing | May 2024 – Sep 2024

- Developed denoising autoencoder for automated imputation of missing client data; used XGBoost for fraud detection.
- Built real-time WebSocket communication system using Python and Flask-SocketIO with asynchronous event handling.
- Engineered cross-platform C modules implementing AES, DES, and Base64 encryption.

PROJECTS

AlphaForge-MAA: AI-Driven Alpha Factor Mining | RL (D3QN/PPO), 3GAN, PyTorch, Qlib

- Replaced linear factor combination with RL agents using IC/ICIR rewards to dynamically adjust weights for market adaptability.
- Integrated 3GAN to generate diverse alpha factors and filter high-IC candidates, boosting multi-factor robustness.
- Built RL trading simulators to train factor rebalancing strategies, merging deep RL with quantitative trading logic.

SKILLS

Languages: C, C++, Python, JavaScript (jQuery), SQL, MATLAB, Bash, Verilog, HTML5, CSS3

ML/AI: PyTorch, TensorFlow, Hugging Face, scikit-learn, Optuna, RAG, LangChain, Dify, LSTM, Q-Learning

DevOps & Tools: Git, Docker, Kubernetes, Jenkins, Argo Workflows, AWS (EC2), GCP (GKE), Linux, CI/CD, SWIG

Databases & Frameworks: MySQL, PostgreSQL, Milvus, Elasticsearch, Flask, FastAPI